

medDreamer: Model-Based Reinforcement Learning with Latent Imagination on Complex EHRs for Clinical Decision Support

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RL for Clinical Decision-Making

Clinical Decision-Making

- **Task:** Timely and personalized treatment decisions in healthcare settings.
- **Difficulty:** High complexity and heterogeneity of physiological responses challenge standardized care protocols.

Reinforcement Learning (RL) vs. Supervised Learning (BC)

- **BC:** mimic historical clinician actions; cannot surpass suboptimal clinical practices.
- **RL:** learn adaptive treatment strategies; mirror the sequential clinical decision-making process and directly optimize long-term patient outcomes.

Major Challenges in RL with EHRs

1. Informative Missingness

- EHR data: irregular sampling, highly sparse observations.
- Missingness conveys key information about patient conditions and dynamics.
- Coarse regular times with imputations → information loss and modeling bias.

2. Fully Offline

- Clinical RL has to train on retrospective data without real exploration.
- Model-free RL is widely favored in practice for algorithmic simplicity and minimal environment assumptions.
- Model-free RL is tightly coupled with clinicians' practices → insufficient exploration and modeling bias.

Contributions

medDreamer: a Model-based RL framework that addresses the dual challenges.

1. A patient world model.

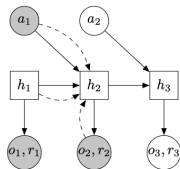
- An Adaptive Feature Integration (AFI) module for handling data irregularity.
- Captures patient dynamics and simulates trajectories via latent imagination.

2. A hybrid two-phase policy training schema.

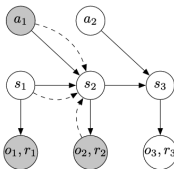
- Trains policies on a blend of real and imagined trajectories.
- Enables safe counterfactual reasoning and rich exploration from purely offline data.

Recap: RSSM & Dreamer

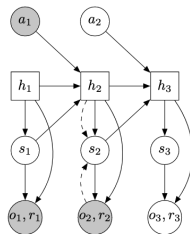
Recurrent State-Space Model (RSSM): split the state into stochastic and deterministic parts.



(a) Deterministic model (RNN)



(b) Stochastic model (SSM)



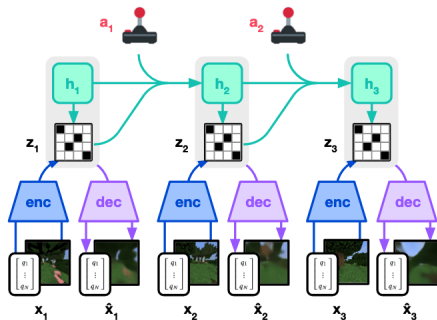
(c) Recurrent state-space model (RSSM)

- The model observes the first two time steps and predicts the third.
- Circles: stochastic variables; squares: deterministic variables.
- Solid lines: generative process; dashed lines: inference model.

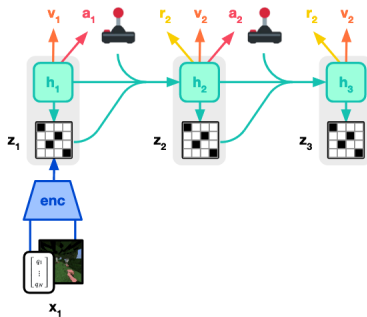
Recap: RSSM & Dreamer

Dreamer:

- 1 World model (RSSM) learns to predict the future based on histories and actions.
- 2 Actor and Critic optimize policies by learning from imagined trajectories.



(a) World Model Learning



(b) Actor Critic Learning

State Space: AFI for Data Sparsity

Adaptive Feature Integration (AFI): treats missingness as a structural signal by modeling both intra-feature temporal dynamics and inter-feature interactions across variables.

For time series $O = (o_1, \dots, o_T) \in \mathbb{R}^{T \times D}$,

$$m_t^d = \begin{cases} 1, & \text{if } o_t^d \text{ is observed} \\ 0, & \text{otherwise} \end{cases} \quad \Delta_t^d = \begin{cases} p_t - p_{t-1} + \Delta_{t-1}^d, & t > 1, m_{t-1}^d = 0 \\ p_t - p_{t-1}, & t > 1, m_{t-1}^d = 1 \\ 0, & t = 1 \end{cases}$$

Project and concatenate to a joint embedding:

$$\mathbf{E}_t^{(o)} = W_o o_t \in \mathbb{R}^{D \times k}, \quad \mathbf{E}_t^{(\Delta)} = W_\delta \Delta_t \in \mathbb{R}^{D \times k},$$

$$\mathbf{E}_t = [\mathbf{E}_t^{(o)} \mid \mathbf{E}_t^{(\Delta)}] \in \mathbb{R}^{D \times 2k}.$$

Apply the Factorization Machine to compute pairwise interactions:

$$\text{FM}(\mathbf{E}_t) = \frac{1}{2} \left(\left(\sum_{j=1}^D \mathbf{e}_t^d \right)^2 - \sum_{j=1}^D (\mathbf{e}_t^d)^2 \right) \in \mathbb{R}^{2k},$$

$$\tilde{o}_t = \sigma \left(\text{Linear}(o_t) + \text{FM}(\mathbf{E}_t) \right) \in \mathbb{R}^{2k}.$$

Reward Space: Intermittent & Terminal Reward

Intermittent rewards: capture physiological improvements or deterioration.

Sepsis treatment:

$$R_{\text{im}}^{\text{Sep}}(s_t, s_{t+1}) = a_{\text{sep}} \cdot \mathbb{1}[s_{t+1}^{\text{SOFA}} = s_t^{\text{SOFA}} \wedge s_{t+1}^{\text{SOFA}} > 0] \\ + b_{\text{sep}} \cdot (s_{t+1}^{\text{SOFA}} - s_t^{\text{SOFA}}) \\ + c_{\text{sep}} \cdot \tanh(s_{t+1}^{\text{lactate}} - s_t^{\text{lactate}})$$

Mechanical ventilation (MV) management:

$$R_{\text{im}}^{\text{Vent}}(s_t, s_{t+1}) = a_{\text{vent}} \cdot \left(\mathbb{1}_{[94 \leq s_{t+1}^{\text{SpO}_2} \leq 98]} - \frac{1}{2} \cdot \mathbb{1}_{[s_{t+1}^{\text{SpO}_2} < 94 \vee s_{t+1}^{\text{SpO}_2} > 98]} \right) \\ + b_{\text{vent}} \cdot \left(\mathbb{1}_{[70 \leq s_{t+1}^{\text{MBP}} \leq 80]} - \frac{1}{2} \cdot \mathbb{1}_{[s_{t+1}^{\text{MBP}} < 70 \vee s_{t+1}^{\text{MBP}} > 80]} \right)$$

Terminal reward: reflect patient survival.

$$R(s_t, s_{t+1}) = \begin{cases} +R_{\text{ter}}, & \text{if } s_{t+1} \in S_T \cap S_{\text{sur}} \\ -R_{\text{ter}}, & \text{if } s_{t+1} \in S_T \cap S_{\text{dec}} \\ R_{\text{im}}(s_t, s_{t+1}), & \text{otherwise} \end{cases}$$

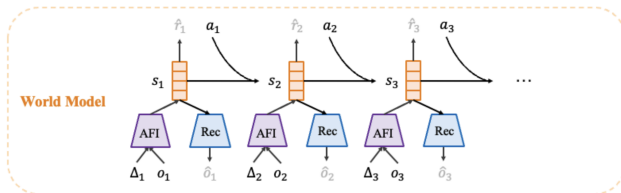
medDreamer: Patient World Model

A. Generative latent dynamics model:

Representation model (posterior): $s_t \sim q_\phi(s_t \mid s_{t-1}, a_{t-1}, \tilde{o}_t)$

Transition model (prior): $\hat{s}_t \sim p_\theta(\hat{s}_t \mid s_{t-1}, a_{t-1})$

- Latent state $s_t \in \mathbb{R}^d$ summarizes the patient history.
- Both q_ϕ and p_θ are parameterized as GRU networks.



Training objective

VAE loss that maximizes the ELBO:

$$\mathcal{L}_D^t \doteq -\text{KL}(q(s_t \mid s_{t-1}, a_{t-1}, \tilde{o}_t) \parallel p(s_t \mid s_{t-1}, a_{t-1}))$$

medDreamer: Patient World Model

B. Auxiliary prediction heads:

Reward model: $\hat{r}_t \sim p_\phi(\hat{r}_t | s_t)$

Continuation model: $\hat{c}_t \sim p_\phi(\hat{c}_t | s_t)$

Reconstruction model: $\hat{o}_t \sim p_\phi(\hat{o}_t | s_t)$.

Training objective

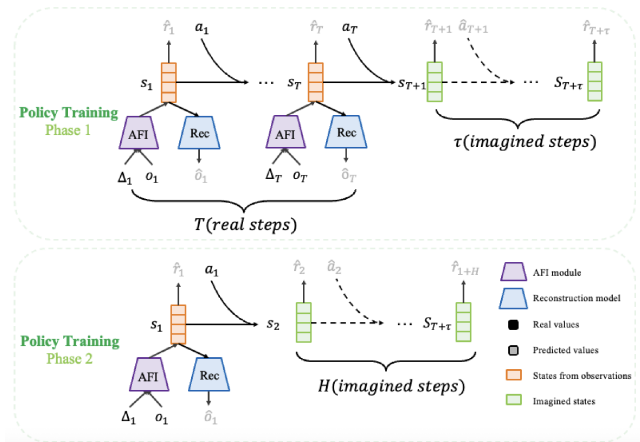
- Standard DreamerV3 loss design:
two-hot symlog $\mathcal{L}_{\text{rew}}$, BCE $\mathcal{L}_{\text{con}}$, symlog MSE $\mathcal{L}_{\text{rec}}$.
- Mask gradient in $\mathcal{L}_{\text{rec}}$ for missing values.

$$\mathcal{L}(\phi) \doteq \mathbb{E}_{q_\phi} \left[\sum_{t=1}^T (\mathcal{L}_{\text{rew}}(\phi) + \mathcal{L}_{\text{con}}(\phi) + \mathcal{L}_{\text{rec}}(\phi) + \mathcal{L}_D(\phi)) \right].$$

medDreamer: Two-Phases Policy Learning

Phase 1: Clinically grounded policy initialization via hybrid states.

Phase 2: Strategic policy refinement through long-horizon imagination.



Imagine 10 steps for phase 1 and 15 steps for phase 2 in experiments.

medDreamer: Actor-Critic Training

- **Bootstrapped λ -return:**

$$v_t = \mathbb{E}[v_\psi(\cdot \mid s_t)],$$

$$R_t^\lambda \doteq \hat{r}_t + \gamma_t \left((1 - \lambda) \cdot v_t + \lambda \cdot R_{t+1}^\lambda \right), \quad R_{T+\tau}^\lambda \doteq v_{T+\tau}$$

- **Actor Loss:**

$$\mathcal{L}_{\text{actor}}(\theta) \doteq - \sum_{t=1}^{T+\tau} \text{sg} \left(R_t^\lambda - V_\psi(s_t) \right) \log \pi_\theta(a_t \mid s_t) + \eta \cdot \mathbb{H}[\pi_\theta(a_t \mid s_t)]$$

- **Critic loss:**

$$\mathcal{L}_{\text{critic}}^{P1}(\psi) \doteq - \sum_{t=1}^{T+\tau} \log p_\psi \left(R_t^\lambda \mid s_t \right)$$

Note: target critic $\bar{V}_\psi$ is updated via EMA as standard Dreamer.

Clinical Decision-Making Tasks

Evaluate medDreameer on two clinical decision-making benchmark tasks:

- **sepsis treatment recommendation:** MIMIC-IV, 21,233 patients, 40 features, 90-day mortality as outcome.
- **mechanical ventilation management:** eICU, 21,595 patients, 41 features, hospital mortality as outcome.

Results

Q1: How well does our policy perform under offline evaluation and mortality estimation?

Table 1: OPE metrics and estimated mortality rates (with 95% CI) for sepsis treatment and MV management on MIMIC and eICU datasets.

Model	Sepsis					MV				
	WIS $\uparrow$	WPDIS $\uparrow$	CWPDIS $\uparrow$	Mortality(%) $\downarrow$	95% CI	WIS $\uparrow$	WPDIS $\uparrow$	CWPDIS $\uparrow$	Mortality(%) $\downarrow$	95% CI
Random	2.17	-0.01	-0.01	–	–	9.93	-0.02	0.09	–	–
Clinician	4.87	0.14	0.09	23.44	[22.12, 24.77]	13.31	0.24	0.18	18.27	[15.48, 21.06]
EZ-Vent[27](BCQ)	10.13	0.02	0.07	18.47	[18.45, 18.49]	7.09	-0.10	0.07	13.78	[13.76, 13.8]
DeepVent[23](CQL)	10.42	0.03	0.07	19.39	[19.38, 19.39]	7.71	-0.10	0.08	13.8	[13.79, 13.8]
DDQN[39]	10.10	0.02	0.07	17.77	[17.75, 17.79]	7.85	-0.10	0.08	21.5	[21.43, 21.56]
MBRL-BNN[40]	4.90	0.00	0.01	16.87	[14.5, 18.53]	6.34	0.03	0.08	16.7	[13.77, 19.62]
MBRL-GMM[26]	-0.74	-0.01	-0.01	17.53	[17.51, 17.55]	4.09	-0.05	0.04	16.8	[16.79, 16.81]
Dreamer[14]	6.85	0.20	0.13	15.37	[13.13, 17.61]	12.95	0.28	0.20	13.96	[13.91, 16.57]
medDreamer	9.25	0.27	0.19	14.25	[14.6, 17.5]	24.06	0.32	0.25	12.44	[11.42, 13.45]

Results

Q2: How does the learned policy align/differ from clinician decisions in treatment strategies?

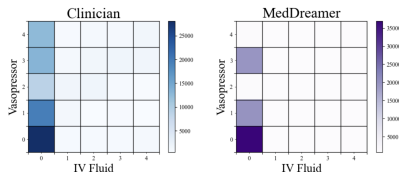


Figure 2: Differences between clinicians' actions and policy actions for sepsis task.

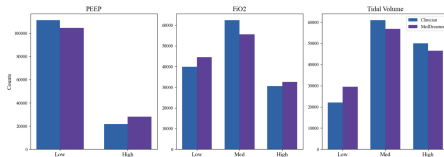


Figure 3: Differences between clinicians' actions and policy actions for MV task.

Results

Q2: How does the learned policy align/differ from clinician decisions in treatment strategies?

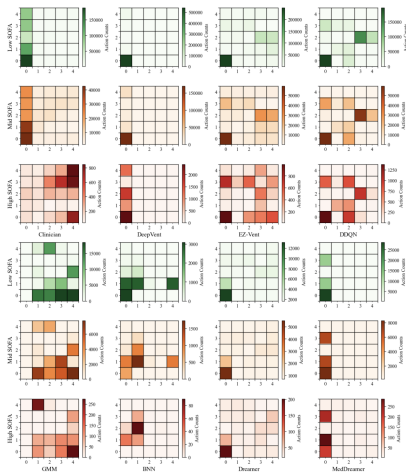


Figure 4: Treatment action distributions for sepsis task.

Results

Q2: How does the learned policy align/differ from clinician decisions in treatment strategies?

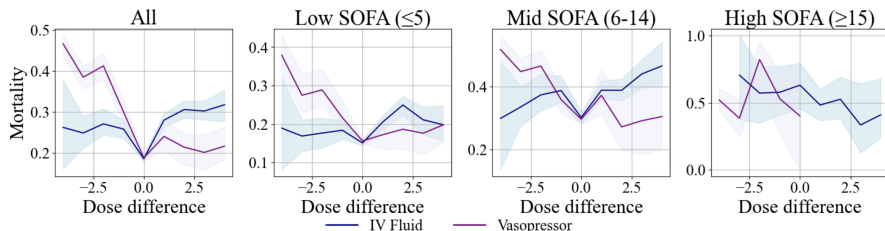


Figure 5: Doses recommended by policy minus clinician vs mortality stratified by SOFA. For patients with high SOFA, there are no cases where policy recommends higher dose of Vasopressor than clinicians.

Results

Q3: Is the learned value function predictive of patient survival?

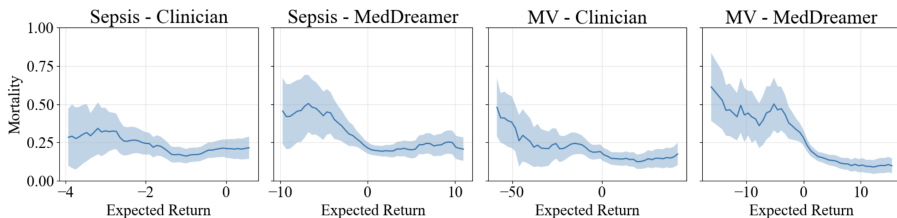


Figure 6: Mortality vs expected return showing medDreamer's ability to assign lower returns to higher risk.

Results

Q4: How do architectural and imagination design choices affect policy performance?

Table 2: Ablations of medDreamer for sepsis treatment.

Model	Sepsis			MV		
	WIS $\uparrow$	WPDIS $\uparrow$	CWPDIS $\uparrow$	WIS $\uparrow$	WPDIS $\uparrow$	CWPDIS $\uparrow$
No AFI	9.10	0.20	0.14	15.77	0.29	0.18
MDP1 Only	7.78	0.19	0.14	17.11	0.29	0.22
MDP2 Only	6.81	0.21	0.13	14.27	0.31	0.22
E2E	10.11	0.21	0.13	8.22	0.26	0.16
ActorCritic	7.65	0.23	0.17	19.81	0.31	0.23
Curriculum	7.65	0.23	0.17	15.93	0.29	0.21
medDreamer($\tau = 5$)	8.34	0.26	0.19	21.99	0.29	0.17
MedDreamer($\tau = 10$)	9.24	0.27	0.19	24.06	0.32	0.25
medDreamer($\tau = 30$)	8.81	0.21	0.15	8.21	0.29	0.18
medDreamer($\tau = 50$)	8.42	0.24	0.17	18.21	0.28	0.17
medDreamer($H = 5$)	8.66	0.24	0.17	16.91	0.29	0.2
MedDreamer($H = 15$)	9.24	0.27	0.19	24.06	0.32	0.25
medDreamer($H = 25$)	9.08	0.26	0.19	16.92	0.28	0.21
medDreamer($H = 35$)	9.01	0.24	0.17	16.15	0.28	0.2
LSTM	8.89	0.22	0.16	16.58	0.29	0.19
Transformer	9.11	0.27	0.19	16.90	0.29	0.20
Random mask	9.17	0.25	0.19	18.16	0.26	0.16

Results

Q5: How reliable is the learned world model in capturing patient physiology and risk?

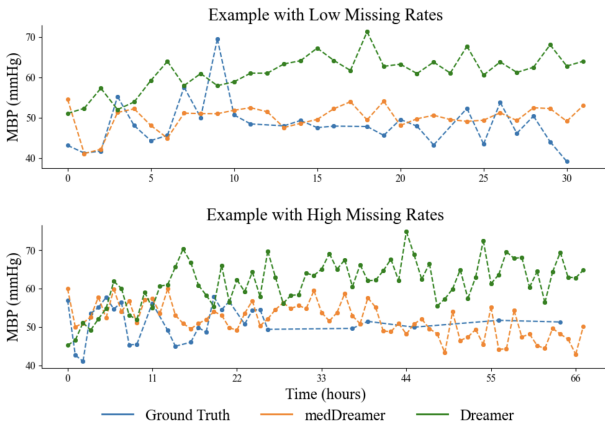


Figure 7: Samples of mean blood pressure trajectories from original dataset, reconstruction of medDreamer and Dreamer with different missing rates.

Results

Q6: How does the policy's expected return evolve during Phases?

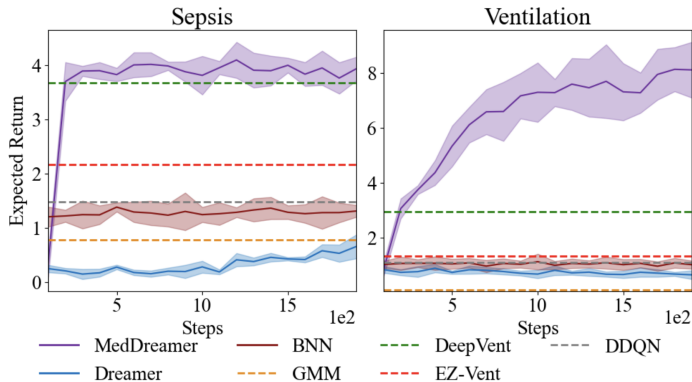


Figure 8: Expected return over training steps.

Conclusion

- Introduced a two-phase model-based RL framework for personalized treatment recommendation for irregular clinical data.
- Extensive experiments on sepsis and ventilation tasks show that medDreamer effectively handles complex EHR data and considerably reduces estimated mortality.

Is it worth reading? Maybe.

- Successfully carry Dreamer-style WM onto EHR data with a systematic evaluation.
- Limited technical novelty beyond DreamerV3.
- Low writing reliability.

Is it worth implementing? Yes.

- Motivate a new way for MBRL in clinical decision-making tasks.